PW SKILLS

Advanced Excel Function: LOOKUP & Statistical Functions

Data Analytics

Learning Objectives

By the end of this session, learners will be able to:

1. Understand and apply lookup functions such as **VLOOKUP, HLOOKUP,** and **XLOOKUP** for efficient data retrieval
2. Use **INDEX** and **MATCH** functions together for flexible and dynamic lookups
3. Perform conditional calculations using **SUMIF** and **COUNTIF** functions
4. Apply **statistical functions** to analyze and summarize data effectively
5. Use **SUMPRODUCT** for advanced multi-condition calculations
6. Handle errors in formulas using IFNA and IFERROR functions
7. Differentiate between various Excel formula categories including **Mathematical, Financial, and Engineering functions**
8. Choose the appropriate function based on real-world data analysis scenarios

Schedule Breakup

Section	Duration
Introduction to Advanced Functions	10 mins
Lookup Functions (VLOOKUP, HLOOKUP, XLOOKUP)	30 mins
INDEX & MATCH (Dynamic Lookup)	25 mins
Conditional Functions (SUMIF, COUNTIF)	20 mins
Break	10 mins
Statistical Functions	20 mins
Advanced Calculations (SUMPRODUCT)	15 mins
Error Handling (IFNA, IFERROR)	15 mins
Other Formula Types (Math, Financial, Engineering)	10 mins
Doubt Resolution	15 mins
Wrap-up	10 mins
Total	**180 mins (3 hrs)**

Introduction to LOOKUP Function

1

The lookup function helps us to retrieve data from a table or range.

2

Common Use Cases: Automating data retrieval, matching data, creating dashboards.

3

Key purposes:

- Locate values based on rows or columns.
- Enable efficient data retrieval.

VLOOKUP : Basics

1

- Syntax: =VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])
  - where lookup_value = value to be found
  - Table_array = table from which data to be retrieved.
  - col_index_num = is the column number in table_array from which matching value should be retrieved.
  - range_lookup = TRUE for approximate match

    FALSE for exact match

2 Searches for a value in the first column of a range.

3 Returns a value in the same row from another column.

VLOOKUP : Basics

Dataset :

	A	B	C	D	E	F	G
1	Product	Category	Price	Units Sold	Region	Revenue	
2	Laptop	Electronics	1000	200	North	200000	
3	Tablet	Electronics	500	150	South	75000	
4	Smartphone	Electronics	800	300	East	240000	
5	Desktop	Electronics	1200	100	West	120000	
6	Monitor	Accessories	300	400	North	120000	
7	Keyboard	Accessories	50	600	South	30000	
8	Mouse	Accessories	20	700	East	14000	
9	Printer	Electronics	150	50	West	7500	
10	Scanner	Electronics	200	40	North	8000	
11	Camera	Electronics	400	80	South	32000	
12							

Lookup Table

Question : Find the units sold for the respective product.

LOOKUP Table

Product	Units Sold
Laptop	
Camera	
Scanner	
Desktop	
Monitor	

Lookup Table

Solution

Lookup Table					
Product	**Units Sold**				
Laptop	=VLOOKUP(J4,$A$1:$F$11,4,FALSE)				
Tablet	150				
Camera	80				
Desktop	100				
Monitor	400				
Scanner	40				

VLOOKUP : Common Errors

1. #N/A: This error occurs when value is not found.

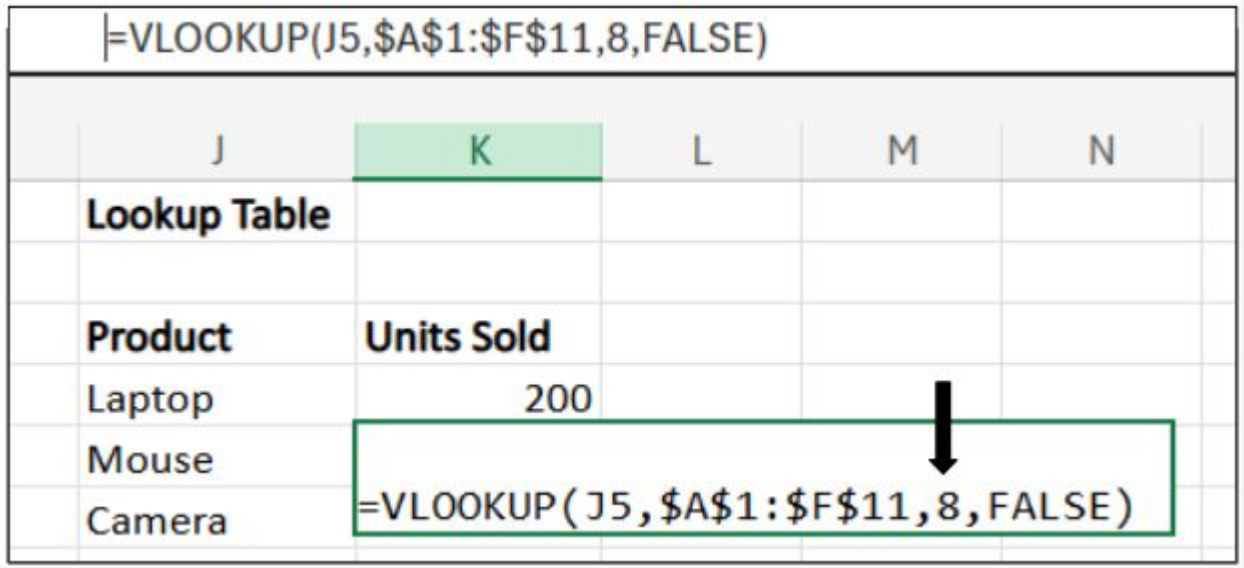
	Product	Units Sold	
	Laptop	200	
	Chair	#N/A	
	Camera	80	

Since in the above table there was not any product named Phone .

2. #VALUE!: Incorrect formula structure gives rise to this error.

VLOOKUP : Common Errors

1. #REF!: Whenever the column index is out of range this error occurs.



Since the column index is out of range i.e. 8.

Necessary Condition

1. **Lookup Value in First Column:** The value you're looking up must be in the first (leftmost) column of the table array.
2. **Valid Column Index:** The column index (col_index_num) must be a positive number within the table's column range.
3. **Lookup Value Exists:** The lookup value should exist in the first column; otherwise, VLOOKUP returns #N/A
4. **Consistent Data Types:** The lookup value and the values in the first column should have the same data type (e.g., both text or both numbers).
5. **Exact or Approximate Match:** Use FALSE (or 0) for an exact match and TRUE (or 1) for an approximate match. When using TRUE, the first column must be sorted in ascending order.

HLOOKUP : Basics

1

- Syntax: =HLOOKUP(lookup_value, table_array, row_index_num, [range_lookup])
  - **lookup_value** = value to be found in the first row of the table
  - **table_array** = table from which data to be retrieved.
  - **row_index_num** = row number in the table from which value is to be returned.
  - **range_lookup** = TRUE for approximate match
    FALSE for exact match

2 Searches for a value in the first row of a range.

3 Returns a value in the same column from another row.

Example Dataset

	A	B	C	D	E	F	G	H	I	J	K
1	**Product**	Laptop	Tablet	Smartphone	Desktop	Monitor	Keyboard	Mouse	Printer	Scanner	Camera
2	**Category**	Electronics	Electronics	Electronics	Electronics	Accessories	Accessories	Accessories	Electronics	Electronics	Electronics
3	**Price**	1000	500	800	1200	300	50	20	150	200	400
4	**Units Sold**	200	150	300	100	400	600	700	50	40	80
5	**Region**	North	South	East	West	North	South	East	West	North	South
6	**Revenue**	200000	75000	240000	120000	120000	30000	14000	7500	8000	32000

Example Dataset

Question : Find the price of the respective product from the lookup table using the sample dataset.

8	**Lookup Table**	
9		
10	**Product**	**Price**
11	Smartphone	
12	Mouse	
13	Camera	
14	Tablet	
15	Monitor	
16	Keyboard	

Example Dataset

Solution :

8	**Lookup Table**		
9			
10	**Product**	**Price**	
11	Smartphone	=HLOOKUP(A11,$A$1:$K$6,3,0)	
12	Mouse		
13	Camera	400	
14	Tablet	500	
15	Monitor	300	
16	Keyboard	50	

Necessary conditions

1. **Lookup Value in First Row:** The value you're searching for must be in the first (topmost) row of the table array.
2. **Valid Row Index:** The row index (row_index_num) must be a positive number within the table's row range.
3. **Consistent Data Types:** The lookup value and the values in the first row should have the same data type (e.g., both text or both numbers).
4. **Lookup Value Exists:** The lookup value should exist in the first row; otherwise, HLOOKUP returns #N/A.
5. **Exact or Approximate Match:** Use FALSE (or 0) for an exact match and TRUE (or 1) for an approximate match. When using TRUE, the first row must be sorted in ascending order.

Limitations of VLOOKUP

Limitation	Description
Static Lookup Position	The lookup value must be in the first (leftmost) column of the table array.
Limited Dynamic Flexibility	Inserting or deleting columns can change the column index, leading to incorrect results.
One-Directional Search	Searches only from **left to right**; it cannot retrieve values from columns to the left.
No Native Multiple Criteria Support	Cannot directly perform lookups using multiple conditions without helper columns or additional functions.
Error for Missing Values	Returns **#N/A** if the lookup value is not found, unless handled with IFERROR or IFNA.

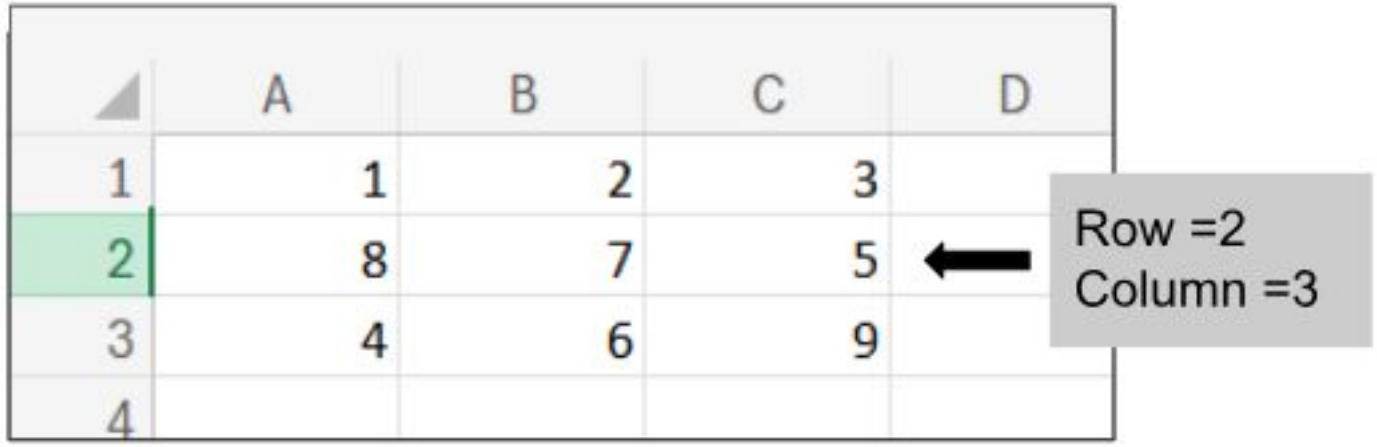
Limitations of HLOOKUP

Limitation	Description
Static Lookup Position	The lookup value must be in the first (topmost) row of the table array.
Limited Dynamic Flexibility	Inserting or deleting rows can change the row index, leading to incorrect results.
One-Directional Search	Searches only from **top to bottom**; it cannot retrieve values from rows above.
No Native Multiple Criteria Support	Cannot directly perform lookups using multiple conditions without helper rows or additional functions.
Error for Missing Values	Returns **#N/A** if the lookup value is not found, unless handled with IFERROR or IFNA.

INDEX Function

- Syntax: =INDEX(array, row_num, [column_num])
- Returns the value of a cell in a specified row and column.

Array



INDEX Function

- Formula : =INDEX(A1:C3,2,3)
- Output : 5

MATCH Function

- **Syntax: =** MATCH(lookup_value, lookup_array, [match_type])
- **Returns the relative position of a value in a range.**

	A	B	C	D
1	1	2	3	
2	8	7	5	
3	4	6	9	
4				

MATCH Function

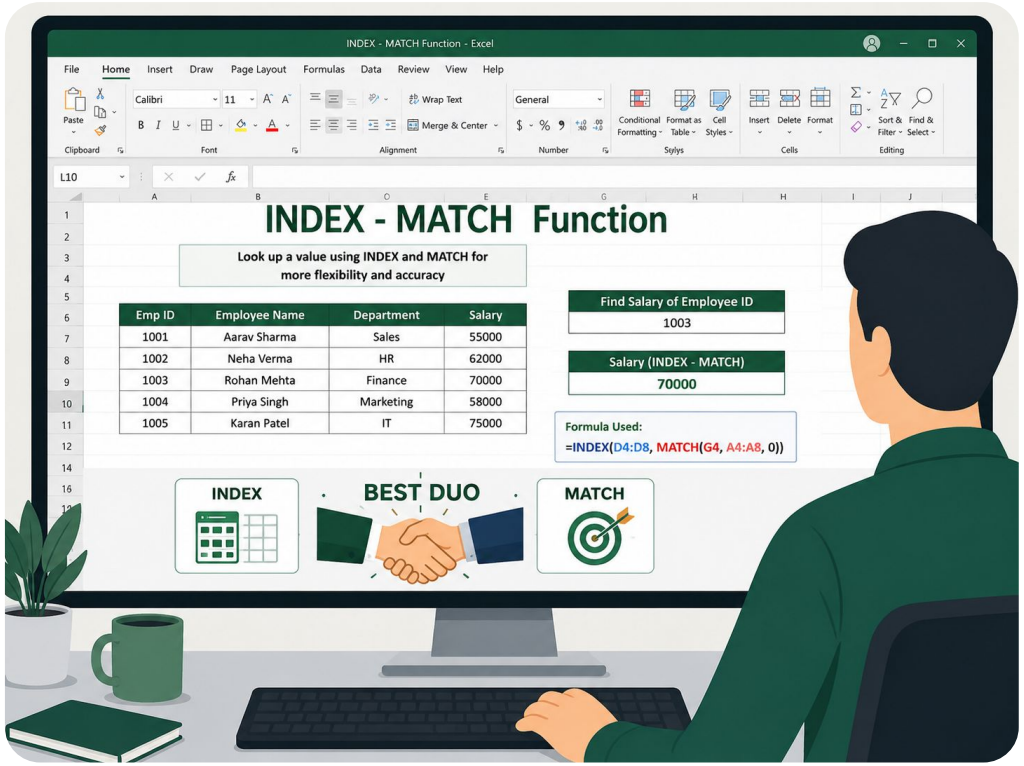
- Formula : = MATCH(8,A2:C2,0)
- Output : 1

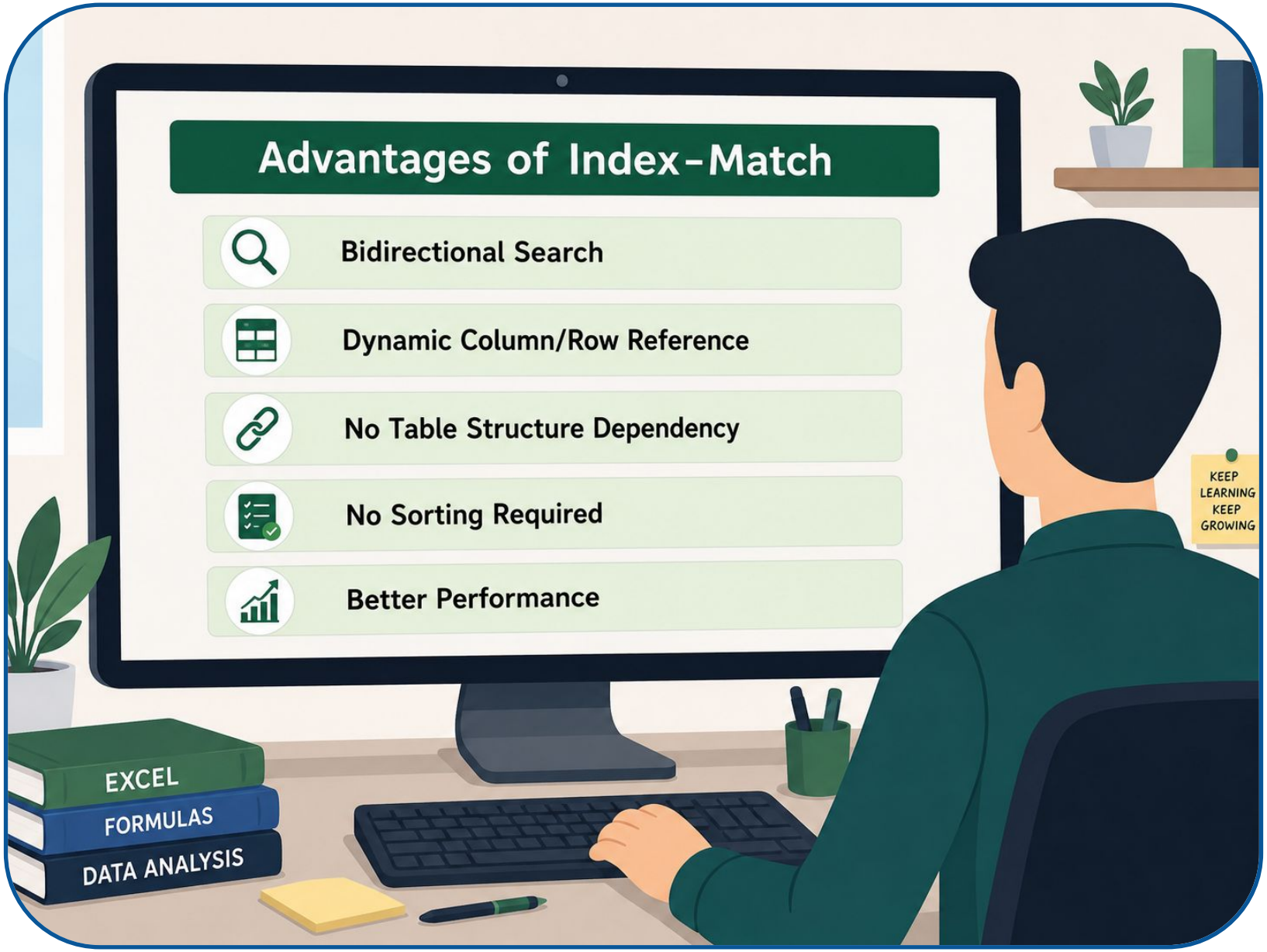
INDEX- MATCH Function

What is INDEX-MATCH?

1. A powerful alternative to VLOOKUP and HLOOKUP.

2. Combines the INDEX and MATCH functions for more flexibility in lookups.




Advantages of Index-Match
Bidirectional Search
Dynamic Column/Row Reference
No Table Structure Dependency
No Sorting Required
Better Performance
KEEP LEARNING KEEP GROWING
EXCEL
FORMULAS
DATA ANALYSIS

INDEX- MATCH Function

PW SKILLS

Example

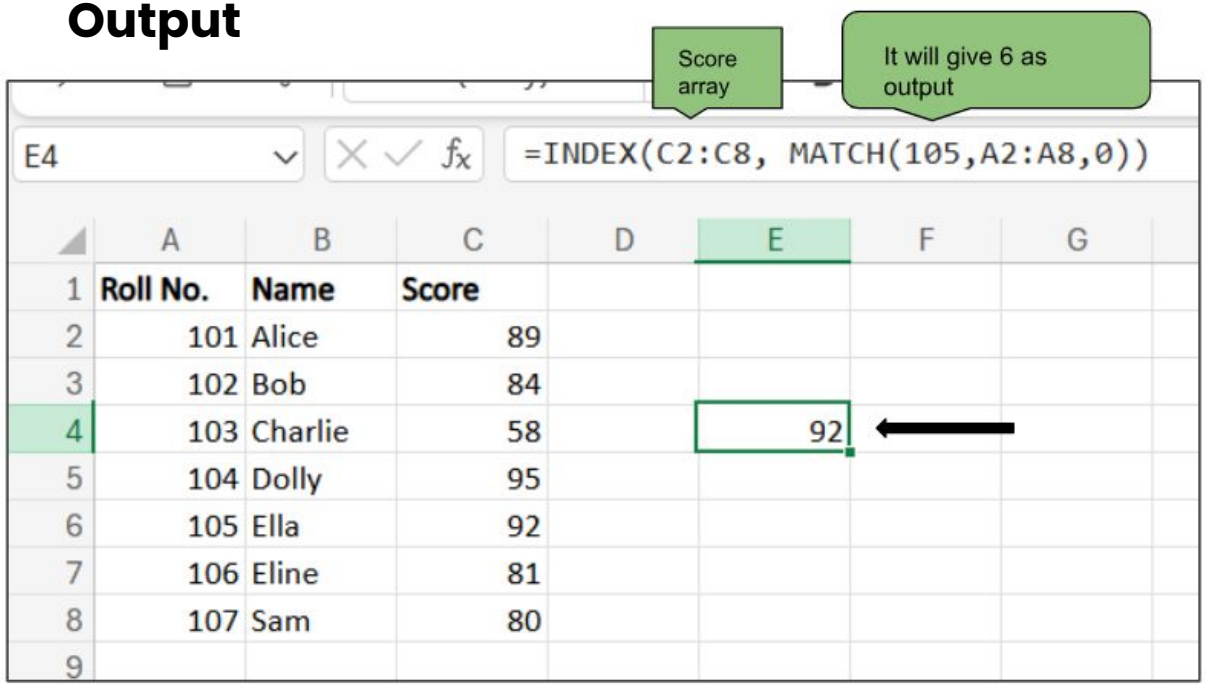
	A	B	C	D	E	F
1	**Roll No.**	**Name**	**Score**			
2	101	Alice	89			
3	102	Bob	84			
4	103	Charlie	74			
5	104	Dolly	95			
6	105	Ella	92			
7	106	Eline	81			
8	107	Sam	80			

We wanted to find the score of Roll No. 105.

INDEX- MATCH Function

Formula : INDEX(C2:C8,MATCH(105,A2:A8,0)

Output



XLOOKUP Overview

Syntax:
=XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found], [match_mode], [search_mode])

- where lookup_value = is the value to search for.
- lookup_array = is the array or range to search for.
- return_array = is the array or range to return.
- if_not_found = returned if match not found
- match_mode = specify how to match lookup_value against the values in lookup_array
- search_mode = specify the search mode you want to use.

Combines the best of VLOOKUP, HLOOKUP, and INDEX-MATCH.

Key Benefits of XLOOKUP

★ Smarter. Simpler. More Powerful.

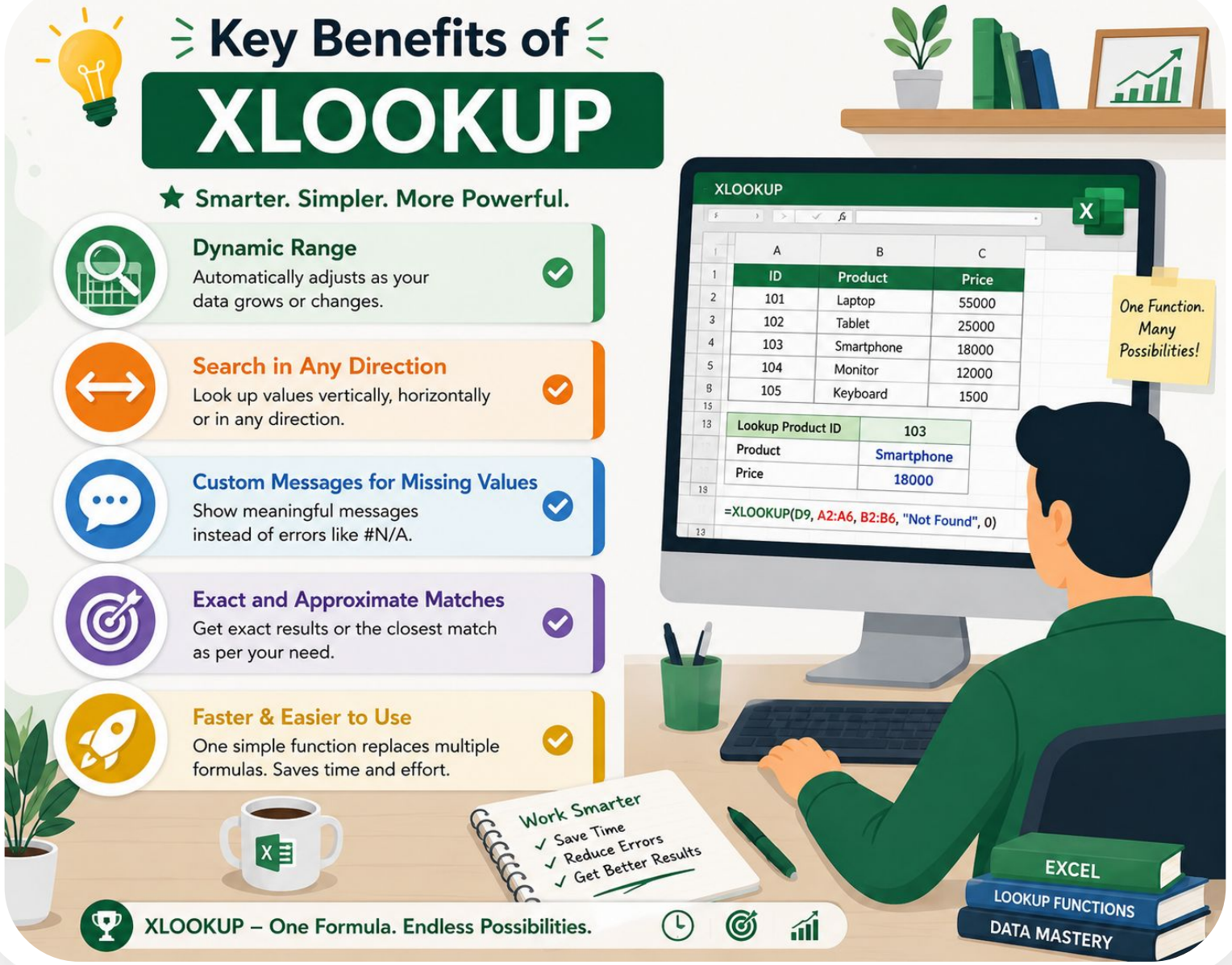
Dynamic Range
Automatically adjusts as your data grows or changes.

Search in Any Direction
Look up values vertically, horizontally or in any direction.

Custom Messages for Missing Values
Show meaningful messages instead of errors like #N/A.

Exact and Approximate Matches
Get exact results or the closest match as per your need.

Faster & Easier to Use
One simple function replaces multiple formulas. Saves time and effort.



XLOOKUP – One Formula. Endless Possibilities.

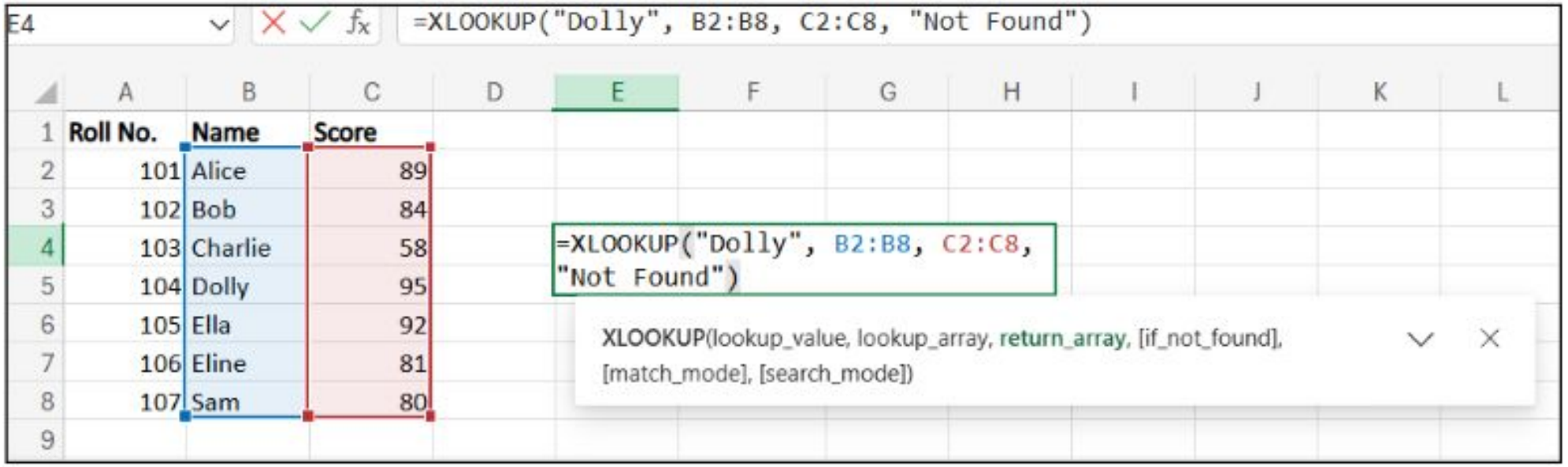
XLOOKUP Example

	A	B	C	D	E	F
1	**Roll No.**	**Name**	**Score**			
2	101	Alice	89			
3	102	Bob	84			
4	103	Charlie	74			
5	104	Dolly	95			
6	105	Ella	92			
7	106	Eline	81			
8	107	Sam	80			

Use Case: Find student details for "Dolly."

SOLUTION

Formula: = XLOOKUP("Dolly", B2:B8, C2:C8, "Not Found")



Comparison of Lookup Functions

Feature	VLOOKUP	HLOOKUP	INDEX-MATCH	XLOOKUP
Leftward Lookup	No	No	Yes	Yes
Dynamic Ranges	No	No	Yes	Yes
Error Handling	No	No	Yes	Yes

REAL-LIFE APPLICATIONS OF

VLOOKUP | **HLOOKUP** | **INDEX-MATCH** | **XLOOKUP**



VLOOKUP

Searches vertically in the first column and returns a value from another column.

REAL-LIFE APPLICATIONS

- **HR & Payroll** — Find employee details using Employee ID.
- **Sales & Orders** — Get product information based on Product ID.
- **Billing & Invoicing** — Retrieve item prices while generating bills.
- **Education** — Look up student grades using Roll Number.

HLOOKUP

Searches horizontally in the first row and returns a value from another row.

REAL-LIFE APPLICATIONS

- **Project Management** — Get project status across different months.
- **Financial Reports** — Fetch quarterly/yearly data from summary tables.
- **Inventory Tracking** — Get stock levels across multiple locations.
- **Sports Analysis** — Look up player statistics across matches.

INDEX-MATCH

Returns a value at the intersection of a specific row and column.

REAL-LIFE APPLICATIONS

- **Data Analysis** — Extract values from large datasets with precision.
- **HR Analytics** — Get department-wise salary information.
- **Performance Tracking** — Analyze KPIs from dynamic reports.
- **Advanced Reporting** — Create flexible reports without column limits.

XLOOKUP

Searches in any direction and returns exact matches with ease.

REAL-LIFE APPLICATIONS

- **Customer Management** — Find customer details by name, phone or ID.
- **Finance** — Lookup transactions, invoice numbers or amounts easily.
- **Data Consolidation** — Pull data from multiple sources quickly.
- **Business Intelligence** — Simplify complex lookups in dashboards.

Look up Smarter.
Analyze Better. Decide Faster.

Interactive Task

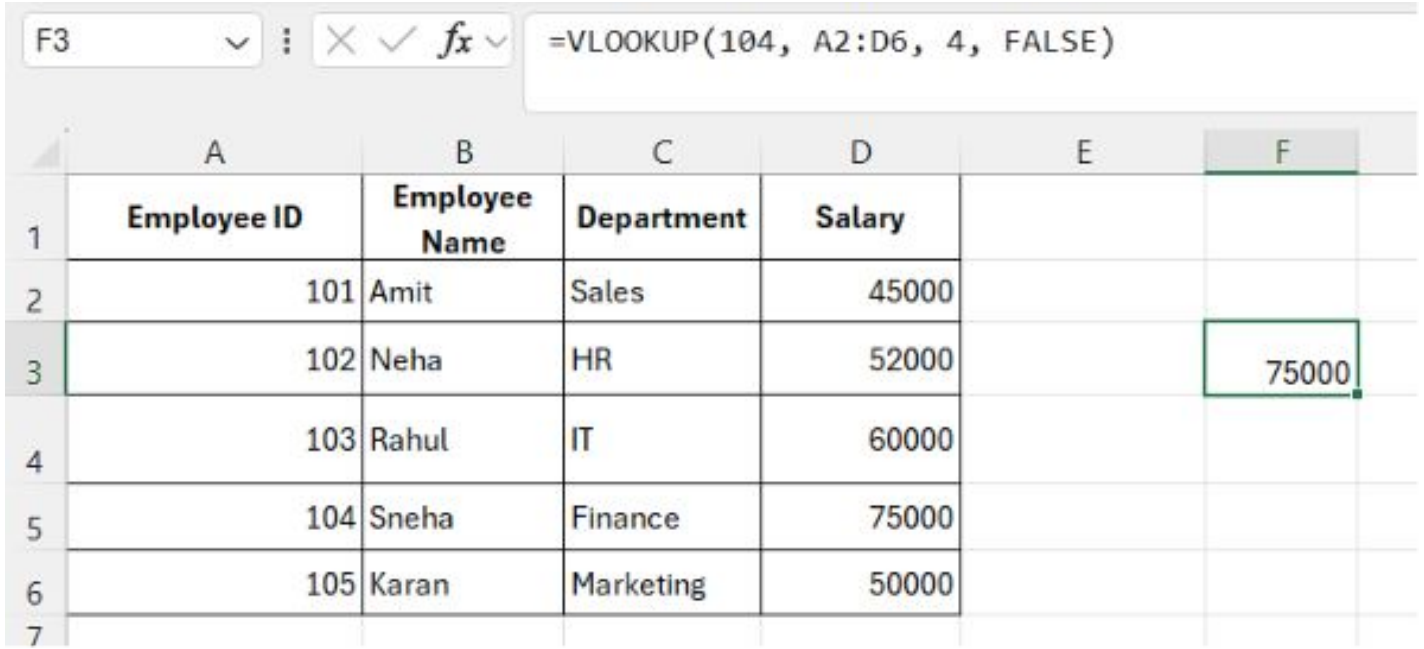
	A	B	C	D
1	**Employee ID**	**Employee Name**	**Department**	**Salary**
2	101	Amit	Sales	45000
3	102	Neha	HR	52000
4	103	Rahul	IT	60000
5	104	Sneha	Finance	75000
6	105	Karan	Marketing	50000

Find the Salary of Employee ID = 104

Interactive Task

Formula: = VLOOKUP(104, A2:D6, 4, FALSE)

Output



Interactive Task

	A	B	C	D
1	**Employee Name**	**Department**	**Employee ID**	**Salary**
2	Amit	Sales	101	45000
3	Neha	HR	102	52000
4	Rahul	IT	103	60000
5	Sneha	Finance	104	75000
6	Karan	Marketing	105	50000

- **What if Employee ID is NOT the first column?**
- **Will VLOOKUP still work?**

But in real-world analysis...

We don't just retrieve data
We also need to:

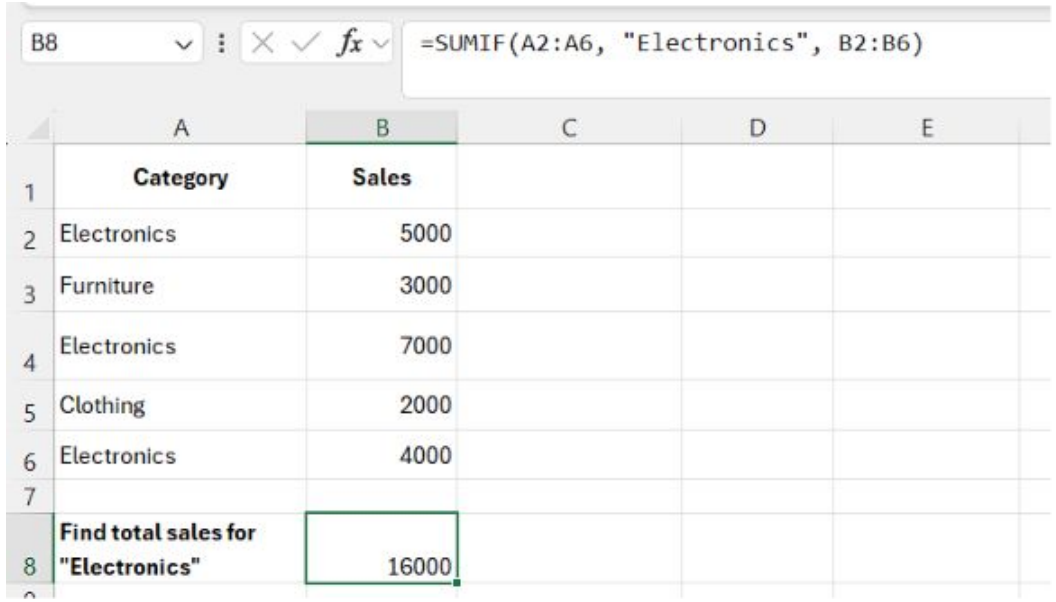
- Calculate totals based on conditions
- Count records meeting criteria
- Summarize data dynamically

Introducing Conditional Functions
SUMIF & COUNTIF

SUMIF Function

Definition

SUMIF is used to **add values based on a specific condition**



	A	B	C	D	E
1	**Category**	**Sales**			
2	Electronics	5000			
3	Furniture	3000			
4	Electronics	7000			
5	Clothing	2000			
6	Electronics	4000			
7					
8	**Find total sales for "Electronics"**	16000			

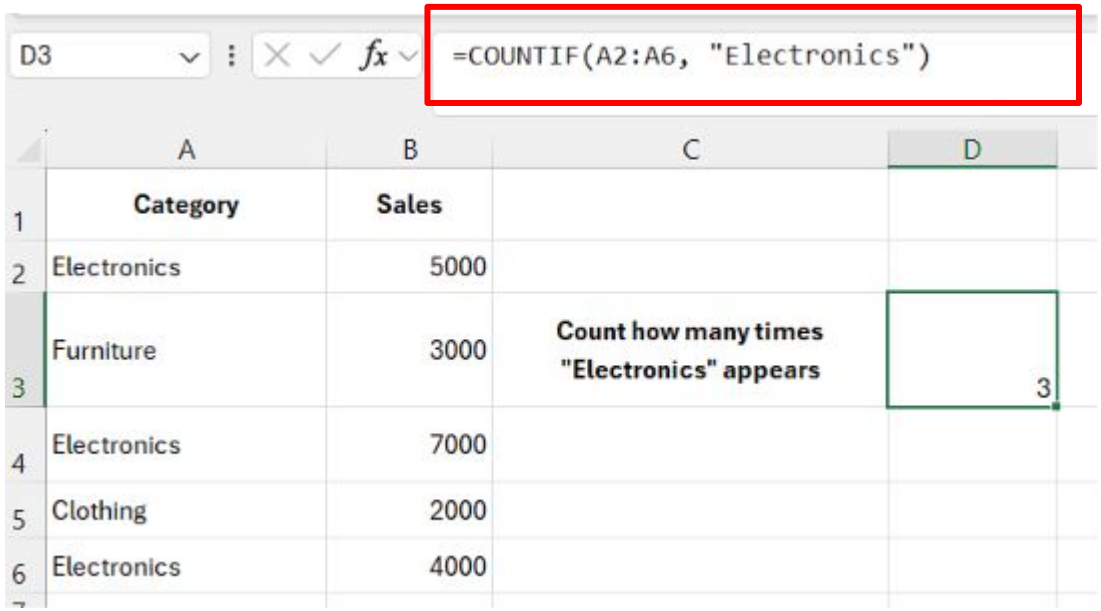
Why it works:

- It checks Category column (A2:A6)
- Matches "Electronics"
- Adds corresponding Sales values (B2:B6)

COUNTIF Function

Definition

COUNTIF is used to **count the number of cells that meet a specific condition**



Explanation:

- A2:A6 → Category range
- "Electronics" → Condition
  Counts matching entries)

Statistical Functions

Used to **analyze data patterns, distribution, and variability**

	Sales
	5000
	3000
	7000
	2000
	4000
	2000
	1400
	4560
MEDIAN → Middle value	3500
MODE → Most frequent value	2000
STDEV.S → Data variation	1886.736

Important Functions

MEDIAN → Middle value
=MEDIAN(A2:A9) → **3500**

MODE → Most frequent value
=MODE.SNGL(A2:A9) → **2000**

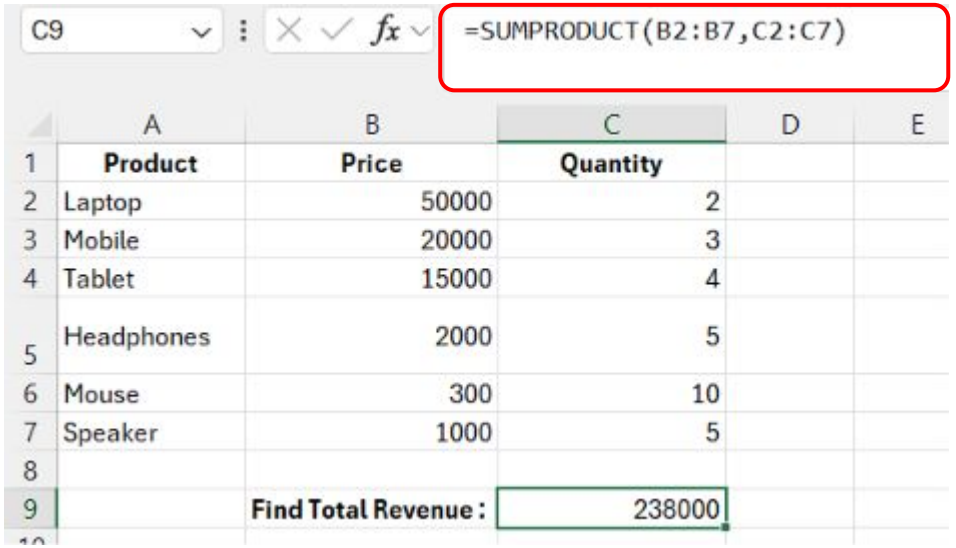
STDEV.S → Data variation
=STDEV.S(A2:A9) → **1886.736**

Why Data Analysts Use This

- Detect outliers
- Understand data spread
- Improve forecasting & reporting

SUMPRODUCT Function

SUMPRODUCT multiplies corresponding values in arrays and returns the **sum of those products**



	A	B	C	D	E
1	**Product**	**Price**	**Quantity**		
2	Laptop	50000	2		
3	Mobile	20000	3		
4	Tablet	15000	4		
5	Headphones	2000	5		
6	Mouse	300	10		
7	Speaker	1000	5		
8					
9		**Find Total Revenue :**	238000		

Why SUMPRODUCT is Powerful

- No need for helper column
- Handles multiple calculations in one formula
- Works like mini data engine

IFERROR

IFERROR FUNCTION : Used for handling errors.

Syntax : **IFERROR**(value,value_if_error)

where value = is any value or expression or reference
value_if_error = value if the error comes

IFERROR

Example :

Name	Age	Gender	Hobby	Assessment Marks	
Alice	25	Female	Reading	85	=IFERROR(B3/C3,"Invalid Calculation")
Bob	22	Male	Cycling	78	
Charlie	28	Male	Painting	92	
Diana	24	Female	Traveling	88	

Name	Age	Gender	Hobby	Assessment Marks	
Alice	25	Female	Reading	85	Invalid Calculation
Bob	22	Male	Cycling	78	
Charlie	28	Male	Painting	92	
Diana	24	Female	Traveling	88	

IFERROR

After using IFERROR

	A	B	C	D	E	F
1	Name	Age	Gender	Hobby	Assessment Marks	
2	Alice	25	Female	Reading	85	
3	Bob	22	Male	Cycling	78	=IFERROR(B3/C3,"Division error")
4	Charlie	28	Male	Painting	92	

	Name	Age	Gender	Hobby	Assessment Marks	
2	Alice	25	Female	Reading	85	
3	Bob	22	Male	Cycling	78	Division error
4	Charlie	28	Male	Painting	92	

IFNA

IFNA FUNCTION : Used for handling #N/A error.

Syntax : IFNA(value , value_if_na)

Example :

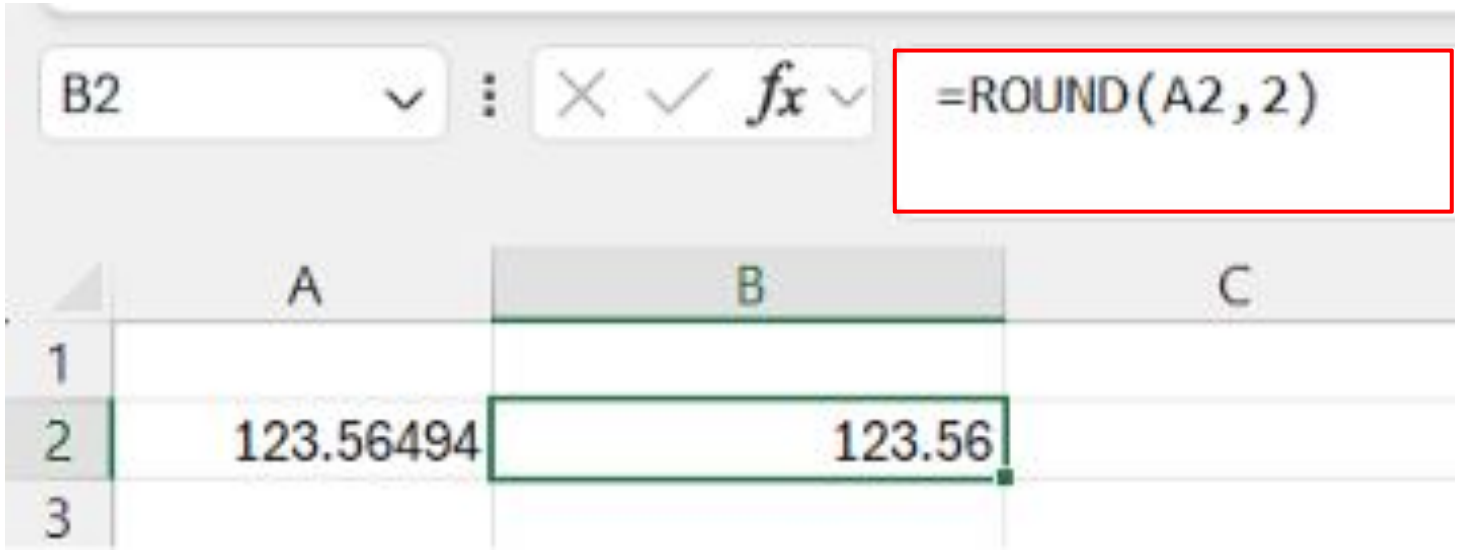
	A	B	C	D	E	F
1	**Name**	**Age**	**Gender**	**Hobby**	**Assessment Marks**	
2	Alice	25	Female	Reading	85	
3	Bob	22	Male	Cycling	78	=IFNA("no #N/A error","error")
4	Charlie	28	Male	Painting	92	

1. Mathematical Functions
(Data Cleaning & Transformation)

Used for **basic to advanced numerical calculations**

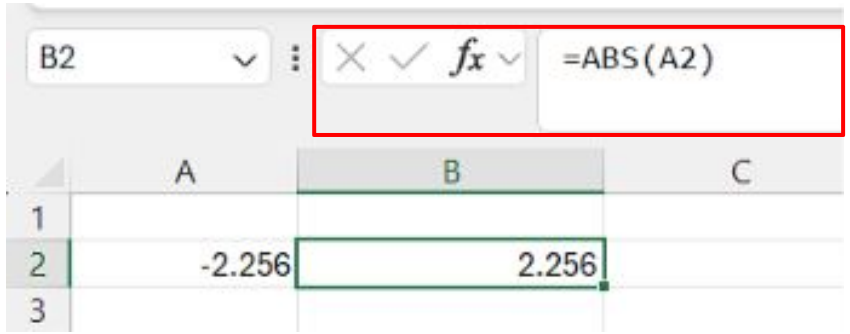
Important Functions

- **ROUND**(number, digits)

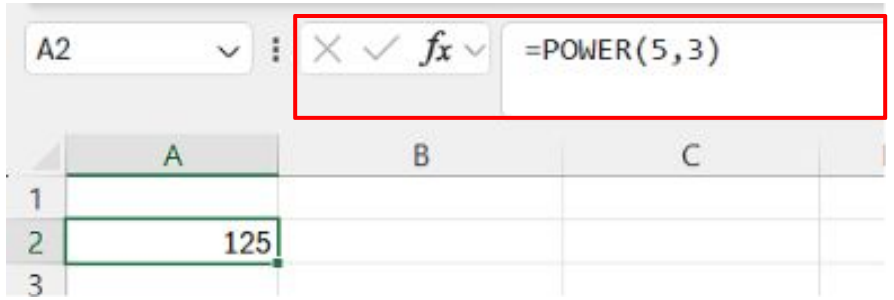


1. Mathematical Functions
(Data Cleaning & Transformation)

- **ABS(number)**
  Converts negative values to positive
  Example: =ABS(-2.256) → 2.256



- **POWER(number, power)**
  Raises number to a power
  Example: =POWER(5, 3) → 125

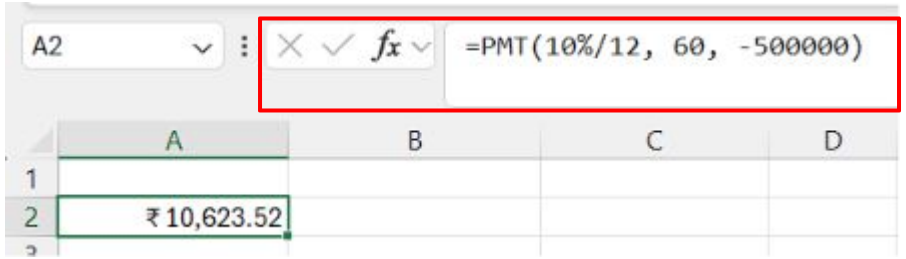


2. Financial Functions
(Business Decision Making)

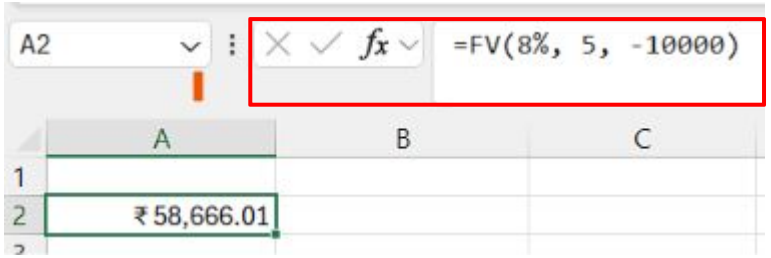
Used for **investment, loan, and financial analysis**

Important Functions

- **PMT(rate, nper, pv)**
  Calculates loan EMI
  Example: =PMT(10%/12, 60, -500000)

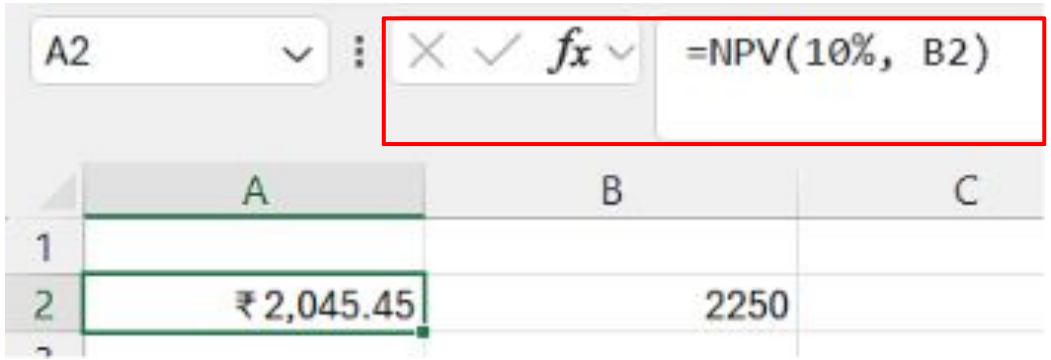


- **FV(rate, nper, pmt)**
  Future value of investment
  Example: =FV(8%, 5, -10000)



2. Financial Functions
(Business Decision Making)

- **NPV(rate, values)**
  Profitability of investment
  Example: =NPV(10%, B2)



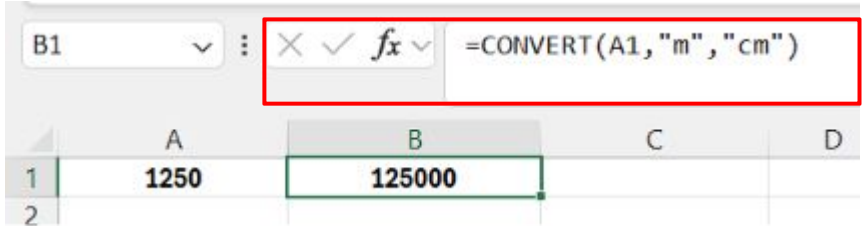
- **Data Analyst Use Case**
  - Loan analysis dashboards
  - Investment comparison models
  - Business forecasting

3. Engineering Functions
(Technical Data Handling)

Used in **scientific, technical, and unit-based calculations**

Important Functions

- **CONVERT(number, from_unit, to_unit)**
  Converts units
  Example: =CONVERT(100, "m", "cm") → 10000



- **DELTA(number1, number2)**
  Checks if two values are equal
  Example: =DELTA(A2, B2)



1

A strong Data Analyst:

- Doesn't memorize all functions
- Understands when to use which category

2

Excel becomes powerful when you:

- Combine different function types
- Apply them to real-world scenarios



Interactive Question

In what scenario would a Data Analyst prefer using **IFNA instead of IFERROR?**



Final Summary

1. **Lookup Functions** → Retrieve data (VLOOKUP, XLOOKUP, INDEX-MATCH)
2. **Conditional Functions** → Analyze with conditions (SUMIF, COUNTIF)
3. **Statistical Functions** → Understand data patterns (MEDIAN, MODE, STDEV.S)
4. **SUMPRODUCT** → Multi-condition calculations
5. **Error Handling** → Clean outputs (IFERROR, IFNA)
6. **Other Functions** → Mathematical | Financial | Engineering

PW SKILLS

Thank You

For Attending the Session

Mastering LOOKUPs is the key to smarter data retrieval.

Keep Learning, Keep Upskilling!